# LIAM FOWL

Postdoctoral Researcher, Princeton Language and Intelligence, Princeton University
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## RESEARCH INTERESTS

My research is on the reasoning of large language models (LLMs): how they learn to reason during training, and how to make their reasoning reliable enough to use in mathematics. I work on AI for mathematics, including natural-language proof generation and formal verification in Lean, and on the learning dynamics of post-training methods such as reinforcement learning and on-policy distillation. My PhD in mathematics was on the security and privacy of machine learning, including data poisoning, backdoors, and privacy attacks on federated learning, and I maintain a strong interest in AI safety.

## APPOINTMENTS

**Postdoctoral Researcher**, Princeton University — Aug 2025 - Present
*Princeton Language and Intelligence; advised by Prof. Sanjeev Arora*

- LLM reasoning for formal (Lean) and informal mathematics.
- Post-training methods: reinforcement learning and on-policy self-distillation.

**Senior Research Scientist**, Google DeepMind — Jun 2022 - Jul 2025
*Gemini, tool use* — Sept 2023 - Jul 2025

- Led the effort to build preference data for Gemini tool use across several extensions; authored production RFCs (including as lead author) and contributed to Gemini's production SFT data mixture.

*Speech Research* — Jun 2022 - Sept 2023

- Led an internationalized federated training routine (custom locale teachers and recipes).
- Primary author of a defensive publication on federated hotword label prompting (dpub #6232).

## EDUCATION

**PhD in Mathematics**, University of Maryland, College Park — 2016 - 2022
Advised by Prof. Wojtek Czaja (Mathematics) and Prof. Tom Goldstein (Computer Science).
Dissertation: *Analysis of Data Security Vulnerabilities in Deep Learning.*

**BS in Mathematics**, University of Maryland, College Park — 2012 - 2016
High honors in Mathematics, minor in Physics.

## HONORS & AWARDS

**Oral presentation**, NeurIPS 2026 (*Escaping the Cognitive Well*) — 2026

**Oral presentation**, CVPR 2022 (*Can Neural Nets Learn the Same Model Twice?*) — 2022

**Ann G. Wylie Dissertation Fellowship**, University of Maryland (top university-wide fellowship) — 2021–2022

**Award for Excellence in Teaching**, University of Maryland (graduate students) — 2017

**Outstanding Senior Award**, top graduating senior in Mathematics, University of Maryland — 2016

**Aziz Mathematics Scholarship**, top undergraduate in Mathematics, University of Maryland — 2016

**Banneker/Key Scholarship**, full scholarship, top 1% of incoming students, University of Maryland — 2012

## PUBLICATIONS

** denotes equal contribution; † denotes equal senior authorship (advising).*

### Conference Papers

[C1] Dang*, X., Agarwal*, R., Porto, R., Goyal, A., **Fowl*†, L.**, & Arora†, S. *Escaping the Cognitive Well: Efficient Competition Math with Off-the-Shelf Models.* Advances in Neural Information Processing Systems **(NeurIPS)** 2026. ***Oral.***

[C2] Kaur, S., Ri, N., He, Y., **Fowl, L.**, & Arora, S. *Rethinking On-Policy Self-Distillation for Thinking Models.* Advances in Neural Information Processing Systems **(NeurIPS)** 2026.

[C3] Lazic, N., **Fowl, L.**, György, A., & Szepesvári, C. *To See the Unseen: On the Generalization Ability of Transformers in Symbolic Reasoning.* Conference on Language Modeling **(COLM)** 2026.

[C4] **Fowl*, L.**, Geiping*, J., Reich, S., Wen, Y., Czaja, W., Goldblum, M., & Goldstein, T. *Decepticons: Corrupted Transformers Breach Privacy in Federated Learning for Language Models.* International Conference on Learning Representations **(ICLR)** 2023.

[C5] Chu, H., Geiping, J., **Fowl, L.**, Goldblum, M., & Goldstein, T. *Panning for Gold in Federated Learning: Targeted Text Extraction under Arbitrarily Large-Scale Aggregation.* International Conference on Learning Representations **(ICLR)** 2023.

[C6] Labrador, B., Zhao, G., Moreno, I. L., Scarpati, A., **Fowl, L.**, & Wang, Q. *Exploring Sequence-to-Sequence Transformer-Transducer Models for Keyword Spotting.* IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** 2023.

[C7] Souri*, H., **Fowl*, L.**, Chellappa, R., Goldblum, M., & Goldstein, T. *Sleeper Agent: Scalable Hidden Trigger Backdoors for Neural Networks Trained from Scratch.* Advances in Neural Information Processing Systems **(NeurIPS)** 2022.

[C8] Wen*, Y., Geiping*, J., **Fowl*, L.**, Goldblum, M., & Goldstein, T. *Fishing for User Data in Large-Batch Federated Learning via Gradient Magnification.* International Conference on Machine Learning **(ICML)** 2022.

[C9] Somepalli, G., **Fowl, L.**, Bansal, A., Yeh-Chiang, P., Dar, Y., Baraniuk, R., Goldblum, M., & Goldstein, T. *Can Neural Nets Learn the Same Model Twice? Investigating Reproducibility and Double Descent From the Decision Boundary Perspective.* Conference on Computer Vision and Pattern Recognition **(CVPR)** 2022. ***Oral.***

[C10] **Fowl*, L.**, Geiping*, J., Czaja, W., Goldblum, M., & Goldstein, T. *Robbing the FED: Directly Obtaining Private Data in Federated Learning with Modified Models.* International Conference on Learning Representations **(ICLR)** 2022.

[C11] **Fowl*, L.**, Goldblum*, M., Chiang, P., Geiping, J., Czaja, W., & Goldstein, T. *Adversarial Examples Make Strong Poisons.* Advances in Neural Information Processing Systems **(NeurIPS)** 2021.

[C12] Geiping*, J., **Fowl*, L.**, Huang, W. R., Czaja, W., Taylor, G., Moeller, M., & Goldstein, T. *Witches' Brew: Industrial Scale Data Poisoning via Gradient Matching.* International Conference on Learning Representations **(ICLR)** 2021.

[C13] Borgnia*, E., Cherepanova*, V., **Fowl*, L.**, Ghiasi*, A., Geiping*, J., Goldblum*, M., . . . , & Gupta*, A. *Strong Data Augmentation Sanitizes Poisoning and Backdoor Attacks Without an Accuracy Tradeoff.* IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** 2021.

[C14] Huang*, W. R., Geiping*, J., **Fowl, L.**, Taylor, G., & Goldstein, T. *MetaPoison: Practical General-Purpose Clean-Label Data Poisoning.* Advances in Neural Information Processing Systems **(NeurIPS)** 2020.

[C15] Goldblum*, M., **Fowl*, L.**, & Goldstein, T. *Adversarially Robust Few-Shot Learning: A Meta-Learning Approach.* Advances in Neural Information Processing Systems **(NeurIPS)** 2020.

[C16] Goldblum, M., Reich*, S., **Fowl*, L.**, Ni*, R., Cherepanova*, V., & Goldstein, T. *Unraveling Meta-Learning: Understanding Feature Representations for Few-Shot Tasks.* International Conference on Machine Learning **(ICML)** 2020.

[C17] Goldblum*, M., **Fowl*, L.**, Feizi, S., & Goldstein, T. *Adversarially Robust Distillation.* AAAI Conference on Artificial Intelligence **(AAAI)** 2020.

[C18] Abdelkader*, A., Curry*, M. J., **Fowl*, L.**, Goldstein*, T., Schwarzschild*, A., Shu*, M., ..., & Zhu*, C. *Headless Horseman: Adversarial Attacks on Transfer Learning Models.* IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP)** 2020. *Spotlight.*

[C19] Goldblum, M., **Fowl, L.**, & Czaja, W. *Sheared Multi-Scale Weight Sharing for Multi-Spectral Superresolution.* Algorithms, Technologies, and Applications for Multispectral and Hyperspectral Imagery XXV, Proc. **SPIE** 10986, 2019.

### Workshop Papers

[W1] Chung*, J., Cai*, Z., Li, Z., Yin, Q., Agarwal, R., Park, S., Porto, R., Ri, N., ..., **Fowl*†, L.**, & Arora†, S. *Goedel-Architect: Streamlining Formal Theorem Proving with Blueprint Generation and Refinement.* AI for Math Workshop, **ICML** 2026; *under review* at **ICLR** 2027.

[W2] He, Y., Kaur, S., Bhaskar, A., Yang, Y., Liu, J., Ri, N., **Fowl, L.**, Panigrahi, A., Chen, D., & Arora, S. *Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision.* AI for Math Workshop, **ICML** 2026 *(Spotlight)*; RLxF Workshop, **ICML** 2026.

[W3] Kalra, N., Ri, N., Bukhari, Z., Li, A., Lotfi, S., **Fowl, L.**, & Goldblum, M. *Measuring the Limits of Continual Learning for LLMs.* Workshop on Compositional Learning, **ICML** 2026.

[W4] **Fowl*, L.**, Chiang*, P., Goldblum*, M., Geiping, J., Bansal, A., Czaja, W., & Goldstein, T. *Preventing Unauthorized Use of Proprietary Data: Poisoning for Secure Dataset Release.* Workshop on Dataset Curation and Security, **NeurIPS** 2020.

[W5] **Fowl*, L.**, Goldblum*, M., Gupta, A., Sharaf, A., & Goldstein, T. *Random Network Distillation as a Diversity Metric for Both Image and Text Generation.* Workshop on Dataset Curation and Security, **NeurIPS** 2020.

[W6] Peri, N., Gupta, N., Huang, W. R., **Fowl, L.**, Zhu, C., Feizi, S., ..., & Dickerson, J. P. *Deep k-NN Defense against Clean-Label Data Poisoning Attacks.* Workshop on Adversarial Robustness in the Real World, **ECCV** 2020.

### Under Review, Preprints, and In Preparation

[P1] **Fowl, L. H.**, Ortega, T., Khaitan, A., Chen, D., Kontorovich, A., & Arora, S. *Goedel-Scribe: Toward Faithful Statement Autoformalization via Adversarial Mutation.* Under review, International Conference on Learning Representations **(ICLR)** 2027; under review, Workshop on Mathematical Reasoning and AI (MATHAI), **NeurIPS** 2026.

[P2] Khaitan, A., **Fowl, L. H.**, Ortega, T., Kontorovich, A., & Arora, S. *Does a Frontier Large Language Model Know What It Can Solve? A 100-Conjecture Calibration Study.* Under review, Workshop on Mathematical Reasoning and AI (MATHAI), **NeurIPS** 2026.

[P3] Khaitan, A., **Fowl, L. H.**, Ortega, T., Kontorovich, A., & Arora, S. *Improving Semantic Equivalence in Research-Level Math Formalization.* Under review, Workshop on Mathematical Reasoning and AI (MATHAI), **NeurIPS** 2026.

[P4] Ortega, T., Park, S., Khaitan, A., **Fowl, L. H.**, Kontorovich, A., & Arora, S. *Do Theorem Provers Need Theorem-Proving Agents? mini-Lean-agent: A Coding-Agent Baseline for Lean.* Under review, Workshop on Mathematical Reasoning and AI (MATHAI), **NeurIPS** 2026.

[P5] **Fowl, L.**, Krishnan, M., Yin, Q., Monson, N., Strauss, H., Arora, S., & Razin, N. *What Can Your Policy See? Investigating Reward Visibility in RLVR.* In preparation.

[P6] Strauss, H., **Fowl, L.**, & Razin, N. *Learning Dynamics of On-Policy Distillation.* In preparation.

[P7] Geiping, J., **Fowl, L.**, Somepalli, G., Goldblum, M., Moeller, M., & Goldstein, T. *What Doesn't Kill You Makes You Robust(er): Adversarial Training against Poisons and Backdoors.* Preprint, arXiv:2102.13624, 2021.

[P8] Huang, W. R., Emam, Z., Goldblum, M., **Fowl, L.**, Terry, J. K., Huang, F., & Goldstein, T. *Understanding Generalization through Visualizations.* Preprint, arXiv:1906.03291, 2019.

[P9] **Fowl, L.**, Partridge, K., Chen, N., & Moreno, I. L. *Improved Labeling for Federated Keyword Spotting Training via Label Prompting.* Defensive publication, dpub #6232.

## INVITED TALKS

**Number Theory Seminar**, Northwestern University 2026

**AI for Math Seminar**, Princeton University 2026

**AI for Math Seminar**, University of Maryland 2026

**Princeton Language and Intelligence Colloquium**, Princeton University 2026

**Research Group Seminar** (host: Prof. Yejin Choi), University of Washington 2025

**LLM Seminar**, Google DeepMind 2025

**Privacy and Security Seminar**, ETH Zürich 2022

## TEACHING & MENTORING

**Courses taught and graded**, University of Maryland 2016 - 2022
Calculus I, II, and III; Statistics; Linear Algebra; Introduction to Machine Learning; Graduate Real Analysis I (grader); Research Interaction Team on mathematical machine learning.

**Students mentored**, Princeton University 2025 - Present

- Xingyu Dang: competition math with off-the-shelf models [C1].
- Rohit Agarwal: competition math and formal theorem proving [C1, W1].
- Rodrigo Porto (undergraduate): competition math and formal theorem proving [C1, W1].
- Jack (Ziyang) Cai: formal theorem proving with blueprints [W1].
- Narutatsu (Edward) Ri: on-policy self-distillation and post-training [C2, W2, W3, W1].
- Helen Hui (undergraduate).

**Mentor**, Directed Reading Program, University of Maryland 2016 - 2018

## SERVICE & OUTREACH

**Reviewer**: ICLR (2020–2026), NeurIPS (2020–2026), CVPR (2021–2026).

**Organizer**, Norbert Wiener Center Seminar, University of Maryland 2018 - 2022

**Organizer**, Deep Learning Research Interaction Team, University of Maryland 2018 - 2022

**Volunteer**, Girls Excelling in Math and Science (GEMS) 2016 - 2018

**Tutor**, Northwestern High School, Prince George's County, MD 2015 - 2018